OpenClaw Temperature and Humidity Acquisition Tutorial

> Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [04-OpenClaw Prompts], [05-AI Voice Interaction Configuration], and [06-Three Conversation Scheme Configuration and Testing] in the `Preparation for Use` directory. If you are currently only using TUI / WhatsApp, you can skip `05` for now. This document assumes the basic environment is already prepared and only expands on DHT-related content.

1. Tutorial Objectives

This tutorial demonstrates OpenClaw's natural language control capabilities through DHT temperature and humidity sensors. After completing this tutorial, readers will be able to:

- Understand the basic idea of OpenClaw reading temperature and humidity
- Interact with OpenClaw via three methods: WhatsApp, TUI, and voice code
- Complete four DHT cases: single reading, comfort judgment, threshold alert, OLED linkage display
- Understand the complete pipeline from "user question" to "sensor reading" and "result feedback"

2. Hardware Preparation

- OpenClaw main board
- DHT temperature and humidity sensor `[Default DHT11]`
- `[Optional] OLED display`
- `[Optional] Microphone module`
- Wiring diagram

3. Software Preparation

Code entry points used in the tutorial

- Low-level driver code: `/home/sunrise/hardware_hal_demo/`
- Temperature and humidity reading command: `/home/sunrise/hardware_hal_demo/dht`
- Temperature and humidity linkage display: `/home/sunrise/hardware_hal_demo/dht_oled`

- Voice entry point: `/home/sunrise/voice_to_openclaw/src/main.py`

The `dht` tool already supports the following capabilities:

- Read current temperature and humidity
- Output results in `json` or `text` format
- Use `mock` fake data for local debugging

```
dht_oled  # Display temperature and humidity data to OLED
dht # Feedback `json` result
```

If you want to display temperature and humidity results directly on the OLED, you can also use `dht_oled`.

If you have selected the AI voice module + speaker, Feishu dialogue, tui, and WhatsApp dialogue need to broadcast the content of openclaw's reply. You can run the following program on the terminal. The AI voice module dialogue does not need to be run repeatedly

4. Testing Temperature and Humidity Acquisition

Before starting this tutorial, please complete a basic `openclaw tui` conversation self-check first; if you haven't done so, see [06-Three Conversation Scheme Configuration and Testing]. After confirming that OpenClaw can respond normally, proceed to the DHT-specific test.

Enter the following command in the terminal to enter TUI:

```
openclaw tui
```

You can try entering the following content in the terminal:

- Read current temperature and humidity
- What is the current environmental humidity?
- Help me judge if the current environment is comfortable
- Remind me if the temperature is too high

```
Read current temperature and humidity

Current temperature and humidity readings:

- Temperature: 29.9°C (85.8°F)
- Humidity: 52%

Sensor: DHT11 ✅

What is the current environmental humidity

Current environmental humidity: 52%

(Temperature is 29.9°C if you need that too)
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
48k/1.0m (5%)
```

5. Three Schemes for Conversing with OpenClaw

Scheme	Suitable Scenarios	Advantages	Suggested Positioning
WhatsApp	Remote control, classroom demonstration	Easy access, suitable for multi-user experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	Get the control pipeline running first
Voice Code	Voice interaction, complete AI experience	Closest to real conversation experience	As the final demonstration scheme

If you have selected the AI voice module + speaker, and WhatsApp conversation, TUI, or WhatsApp conversation need to broadcast OpenClaw's reply content, you can run the following program in the terminal. AI voice module conversation does not need to be run repeatedly.

- Modify whether to enable broadcast parameter

```
/home/sunrise/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  # This parameter controls whether to broadcast. After modifying and saving, run the following program in the terminal.
```

- Enter in terminal:

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

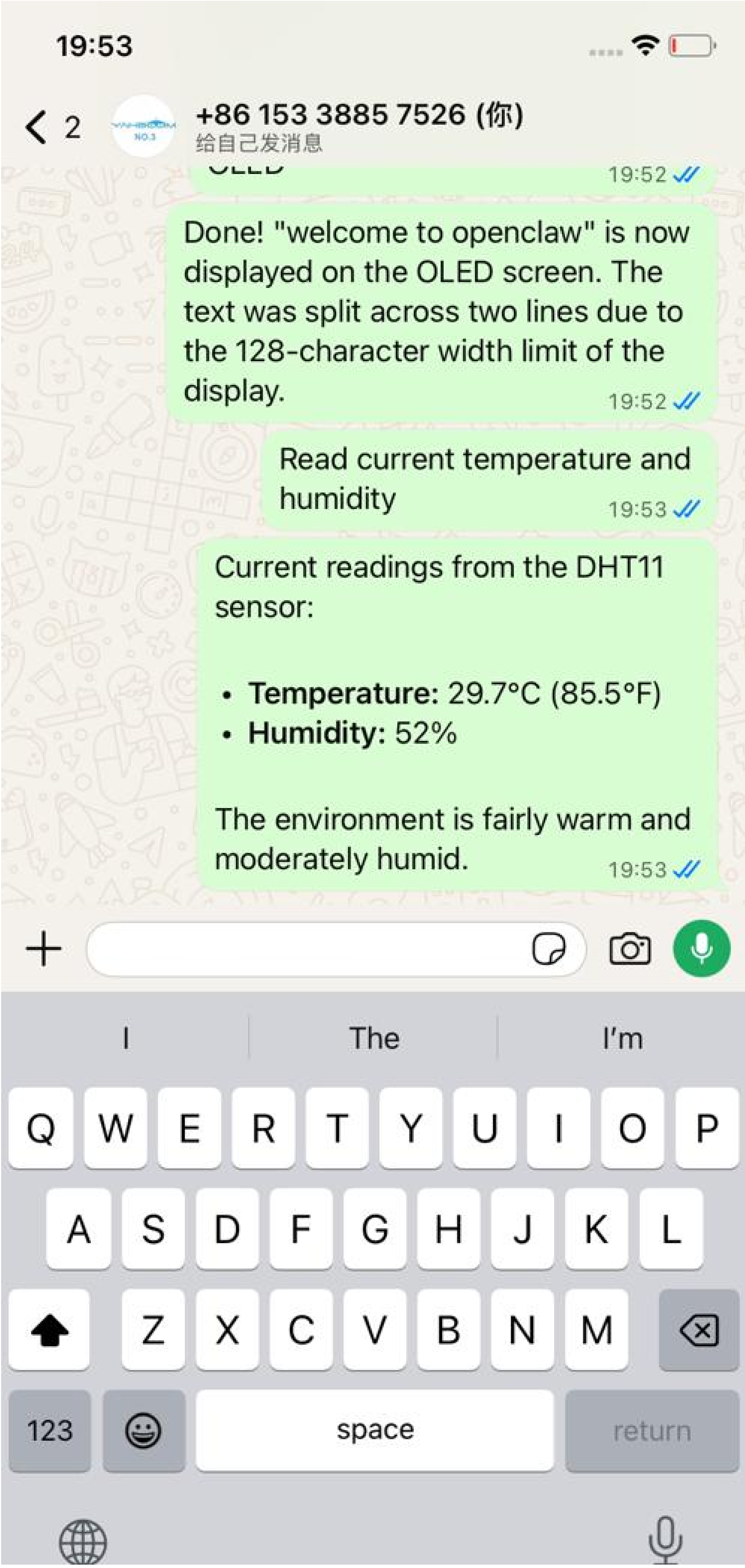
```
(.venv) sunrise@ubuntu:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/sunrise/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp Scheme

For WhatsApp access steps, please refer directly to [06-Three Conversation Scheme Configuration and Testing]. This section only retains conversation examples suitable for the DHT tutorial.

After creation, you can converse directly, for example:

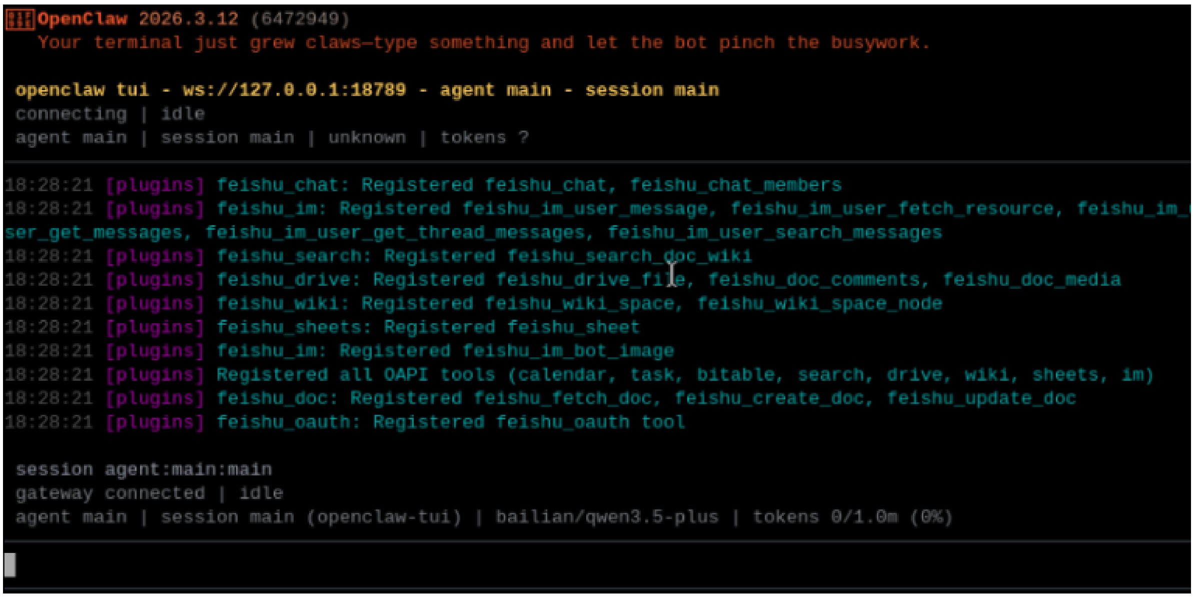
- Read current temperature and humidity
- Judge if it is a bit hot now
- Remind me if the humidity is too low


19:53
+86 153 3885 7526 (你)
给自己发消息
19:52
Done! "welcome to openclaw" is now displayed on the OLED screen. The text was split across two lines due to the 128-character width limit of the display.
19:52
Read current temperature and humidity
19:53
Current readings from the DHT11 sensor:
• Temperature: 29.7°C (85.5°F)
• Humidity: 52%
The environment is fairly warm and moderately humid.
19:53
I
The
I'm
space
return

5.2 TUI Scheme

If you have completed the TUI self-check in Part 6, you can continue the conversation by entering the following command in the terminal:

```
openclaw tui
```



5.3 Voice Code Scheme

```
Requires selection of AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please refer directly to [05-AI Voice Interaction Configuration].

You need to say "Hello Xiao Ya" to wake up and start the conversation.

After wake-up, you can directly speak control content, for example:

- What is the temperature now?
- Help me read the humidity
- Tell me if it is too stuffy

6. Comprehensive Cases

Note: The following are conversation demonstrations conducted via `openclaw tui`. You can also choose WhatsApp or voice methods to complete them.

6.1 Case 1: Single Reading

Experiment Objective

Obtain current temperature and humidity once to complete the most basic environmental sampling.

Effect Description

After receiving the instruction, OpenClaw will call the DHT driver to read the current temperature and humidity and return the results to the user in natural language.

Example Instructions

- Read current temperature and humidity
- What is the indoor temperature now?
- Help me check the current humidity

6.2 Case 2: Comfort Judgment

Experiment Objective

Let OpenClaw make a simple judgment on the current environment after reading the data.

Effect Description

In addition to directly reporting numbers, OpenClaw can also roughly judge states such as "a bit hot," "a bit cold," "a bit dry," or "a bit stuffy" based on temperature and humidity, making the results closer to natural conversation.

Example Instructions

- Help me judge if the current environment is comfortable
- Is the current temperature and humidity suitable for long-term office work?
- Give me a short evaluation based on temperature and humidity

6.3 Case 3: Threshold Alert

Experiment Objective

Let OpenClaw give a reminder when temperature or humidity exceeds the set range.

Effect Description

When the temperature is too high or too low, or when the humidity is too high or too low, OpenClaw can directly output reminder information, or link with RGB lights for more obvious feedback.

Example Instructions

- Remind me if the temperature exceeds 30 degrees
- Prompt that the air is a bit dry if the humidity is below 40%
- Has the alarm condition been reached now?

Expandable Gameplay

- Link with RGB for red, yellow, and blue three-color reminders
- Link with OLED to display alarm text
- Add a buzzer for stronger reminders

6.4 Case 4: OLED Linkage Display

Experiment Objective

Display the just-read temperature and humidity results directly on the OLED, forming a complete data display pipeline.

Effect Description

After completing an environmental sampling, OpenClaw can not only reply the results to the user but also synchronously write the temperature and humidity to the OLED screen, forming a complete effect of "reading + displaying on screen."

Example Instructions

- Read current temperature and humidity and display on OLED
- Write temperature and humidity to the screen
- Make a simple temperature and humidity display page

Expandable Gameplay

- Periodically refresh OLED content
- Make a temperature and humidity dashboard
- Link with the weather station project to form a complete application